

Healthcare Analytics

SQL Case Study

Case Study Overview

A healthcare organization wants to analyze patient records, hospital utilization, treatments, costs, and patient length of stay. The organization maintains related data across patients, hospitals, admissions, and treatments. Your goal is to use SQL to transform this data into meaningful operational and financial insights.

Database Structure

Table	Key fields
Patients	PatientID (Primary Key), FullName, Age, Gender, Address
Hospitals	HospitalID (Primary Key), HospitalName, Location, Capacity
Admissions	AdmissionID (Primary Key), PatientID (Foreign Key), HospitalID (Foreign Key), AdmissionDate, DischargeDate, ReasonForAdmission
Treatments	TreatmentID (Primary Key), AdmissionID (Foreign Key), ProcedureName, Cost, Outcome

Business Questions

- How many patients are there in each gender group, and what is the average age of patients?
- Which hospitals have the highest number of admissions?
- What is the total treatment cost generated by each hospital?
- What is the average length of stay for patients at each hospital?
- Which patients stayed in a hospital for more than 7 days?
- Which treatments have been performed more than 5 times across all hospitals?
- How can admission and treatment information be combined to create complete patient histories?
- How can patients admitted for different reasons, such as surgery and therapy, be combined into a single result?

Advanced Analytics

- **Hospital Cost Analysis:** Use a subquery to identify the hospital with the highest average treatment cost.
- **Hospital Performance:** Create a view named **HospitalPerformance** showing total admissions, average length of stay, and total revenue generated for each hospital.
- **Hospital Revenue Ranking:** Use the **RANK()** window function to rank hospitals according to total revenue.
- **Treatment Frequency Ranking:** Use the **DENSE_RANK()** window function to rank treatments according to how frequently they are performed.

Expected Analysis

The analysis should demonstrate the use of relational database concepts and SQL analytics to answer practical healthcare questions. Queries may involve joins, aggregation functions, filtering, subqueries, views, and window functions.

Recommended SQL Skills

- SELECT, WHERE, AND/OR, and filtering conditions
- INNER JOIN and LEFT JOIN
- GROUP BY and aggregate functions such as COUNT(), SUM(), and AVG()
- Date calculations for length-of-stay analysis
- Subqueries
- CREATE VIEW
- RANK() and DENSE_RANK() window functions

Case Study Objective

Build and analyze the healthcare database to produce clear, accurate insights into patient demographics, hospital utilization, treatment costs, length of stay, hospital performance, revenue, and treatment frequency.